MPJ LAB-1

AIM

Write a Java Prog to showcase classes, object creation, methods, constructors using student information

OBJECTIVES

To study

- classes, objects and creation of objects of class
- methods, return a value from the method, and pass parameter to methods
- constructors and use of constructors
- different access specifiers in Java and their use for encapsulation

THEORY

→ Class

A class is user-defined blueprint or template used to create objects
It defines data members & member functions that describe behaviour of objects
A class does not occupy memory until an object is created

→ Object

It is a real-world entity and an instance of a class. It represent actual values of the variables defined in class and occupies memory.

Explanation

Student → class

S1 → object

name, rollno → Data members

Display() → method

→ Syntax to Define Methods in class

General Syntax of Method

```
returnType methodName(ParameterList) {
    // method body
    return value;
}
```

```
class Calc {
    int add(int a, int b) {
        return a+b;
    }
}
public class Main {
    public static void main(String[] args) {
        Calc c = new Calc();
        int result = c.add(10,20);
        System.out.println("sum = " + result);
    }
}
```



→ Constructor

It is a special method that is automatically called when an object is created

It is used to initialize the data members of class

→ Characteristics of Constructors

- Constructor name is same as class name
- It has no return type
- Called automatically during object creation
- Used for initialization

→ Example for Constructor in Java

```
class Student {
    int rollno;
    Student(int r) {
        rollno = r;
    }
    void display() {
        System.out.println(rollno);
    }
}
public class Main {
    public static void main(String[] args) {
        Student s = new Student(101);
        s.display();
    }
}
```

PLATFORM
Open Source Java Prog Tool like Eclipse Editor / Netbeans

CONCLUSION
Thus studied what are classes object creation, methods, constructors, access specifiers

FAQ

Q1) What is class and object?
→ Class
A class is blueprint or template used to define data members and methods. It does not occupy memory until an object is created

→ Object
A object is an instance of class that represents a real-world entity & occupies memory

Q2) What is Construction in Java?
→ It is a special method in Java that is automatically called when an object is created
It is used to initialize data members of class

Key points
- Name is same as class name
- No return type
- Called auto during object creation



Q3) What is default access specifier and what is its scope?
→ The default access specifier is used when no access specifier is specified
→ Its scope is within the same package only
→ It cannot be accessed outside package

Q4) What are different access specifiers
→ Public - accessible from anywhere

→ Protected - Accessible within the same package and subclasses

→ Default - accessible only within same package

→ Private - accessible only within same class